

1. Objective

The objective of this test was to determine whether **Suricata IDS could detect a Cross-Site Scripting (XSS) attempt** against an OWASP Juice Shop application running in a controlled laboratory environment. The test involved generating a harmless XSS test payload against Juice Shop and monitoring the network traffic with Suricata to determine whether the configured detection rule generated an alert.

2. Lab Environment

Component	Description
Target Application	OWASP Juice Shop
Target OS	Windows
IDS	Suricata 8.0.5
IDS OS	Kali Linux
Virtualization	VMware
Target Port	TCP/3000
Network monitoring interface	eth0
Detection rules	/var/lib/suricata/rules/local.rules

3. Detection Rule

alert http any any -> any any (msg:"XSS Payload"; http.uri; content:"/x"; sid:1000099; rev:1;)

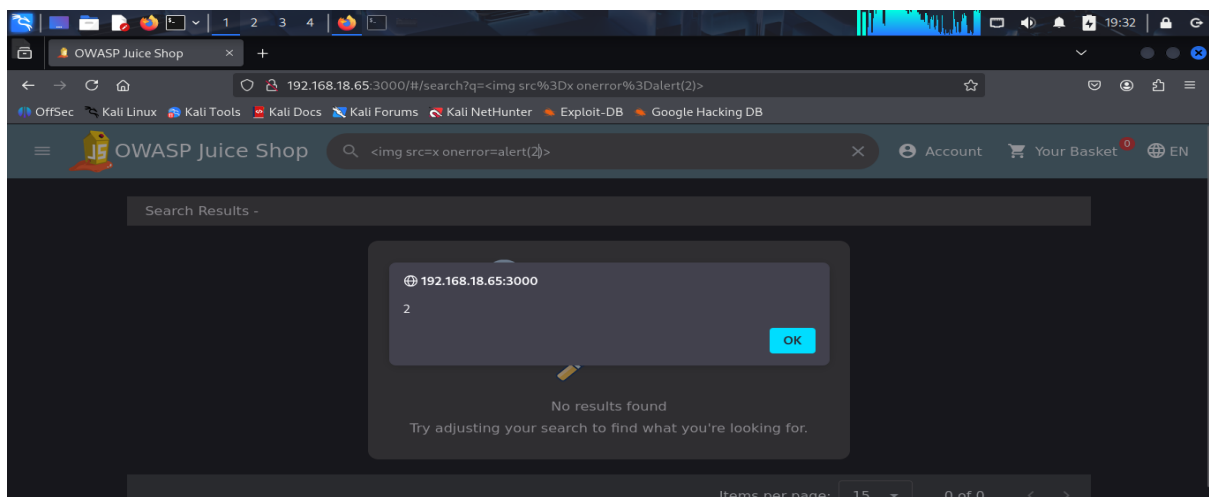
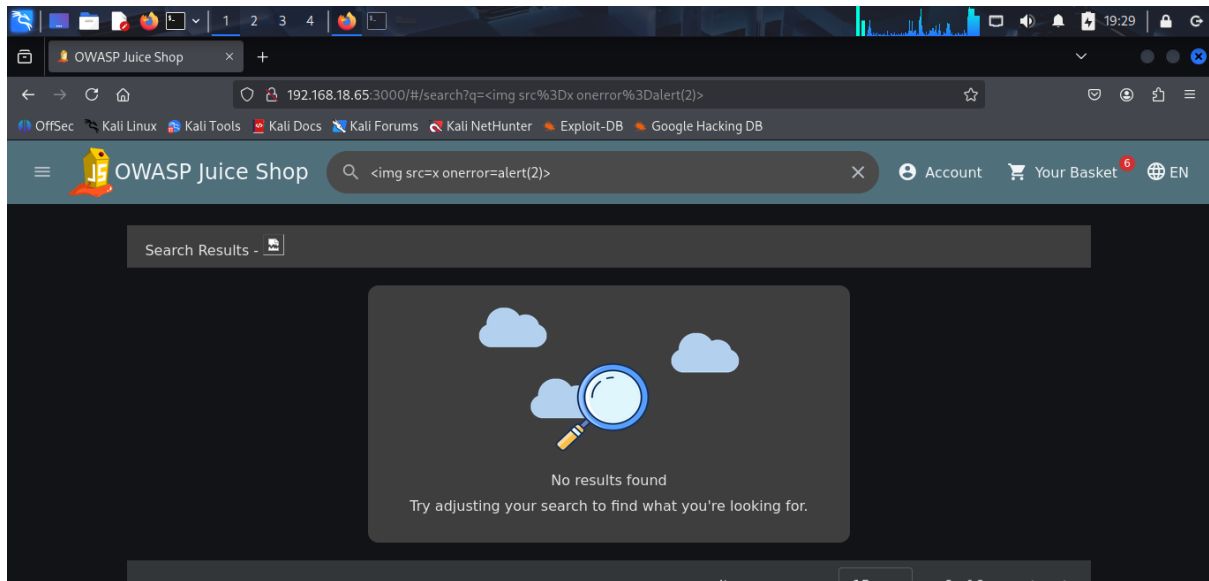
4. Test method

- Typed the XSS script into the Juice Shop search bar and used the fast.log file to captured/receive alerts.
- The following command was used in the terminal to get the fast.log alerts in real time: **sudo tail -f /var/log/suricata/fast.log**

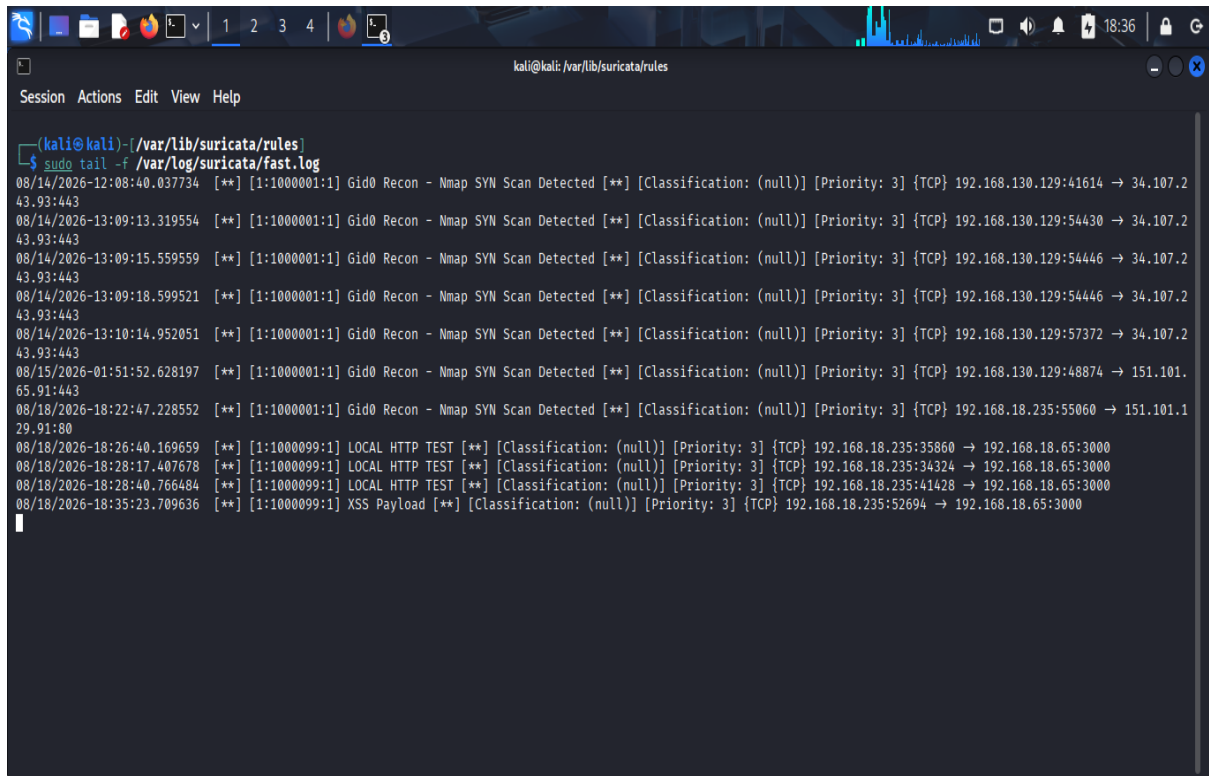
5. Conclusion

The test successfully generated network traffic between the Windows Juice Shop host and the Kali Linux Suricata sensor.

Suricata successfully processed the traffic and generated an alert corresponding to the configured XSS detection rule.



The last four fast.log alerts were created by the payload seen above in the search bar



```
kali@kali: /var/lib/suricata/rules
Session Actions Edit View Help

(kali@kali)~/var/lib/suricata/rules
$ sudo tail -f /var/log/suricata/fast.log
08/14/2026-12:08:40.037734  [**] [1:1000001:1] Gid0 Recon - Nmap SYN Scan Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.130.129:41614 → 34.107.2
43.93:443
08/14/2026-13:09:13.319554  [**] [1:1000001:1] Gid0 Recon - Nmap SYN Scan Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.130.129:54430 → 34.107.2
43.93:443
08/14/2026-13:09:15.559559  [**] [1:1000001:1] Gid0 Recon - Nmap SYN Scan Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.130.129:54446 → 34.107.2
43.93:443
08/14/2026-13:09:18.599521  [**] [1:1000001:1] Gid0 Recon - Nmap SYN Scan Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.130.129:54446 → 34.107.2
43.93:443
08/14/2026-13:10:14.952051  [**] [1:1000001:1] Gid0 Recon - Nmap SYN Scan Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.130.129:57372 → 34.107.2
43.93:443
08/15/2026-01:51:52.628197  [**] [1:1000001:1] Gid0 Recon - Nmap SYN Scan Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.130.129:48874 → 151.101.1
65.91:443
08/18/2026-18:22:47.228552  [**] [1:1000001:1] Gid0 Recon - Nmap SYN Scan Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.18.235:55060 → 151.101.1
29.91:80
08/18/2026-18:26:40.169659  [**] [1:1000099:1] LOCAL HTTP TEST [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.18.235:35860 → 192.168.18.65:3000
08/18/2026-18:28:17.407678  [**] [1:1000099:1] LOCAL HTTP TEST [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.18.235:34324 → 192.168.18.65:3000
08/18/2026-18:28:40.766484  [**] [1:1000099:1] LOCAL HTTP TEST [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.18.235:41428 → 192.168.18.65:3000
08/18/2026-18:35:23.709636  [**] [1:1000099:1] XSS Payload [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.18.235:52694 → 192.168.18.65:3000
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